Operations research: review

Optimization · Level 4 · Unit check



In this unit

- Linear programming and the simplex method
- Integer programs and branch and bound
- Constraint programming: propagation and search
- Heuristics and trade-offs

The big ideas

- LP: corners, simplex and shadow prices
- MIP: relaxations bound, branch and bound proves
- CP: propagate, search fail-first; INFEASIBLE is a proof
- Heuristics: build, improve, measure the gap, show the trade-offs

Activity: Pick the tool

1. Sort (6 min): Match ten problems to LP, MIP, CP or a heuristic.
2. Model (8 min): Write one in a paragraph.
3. Report (6 min): What would you report besides the answer?

Many problems fit more than one tool; the reason matters more than the label.

Let’s discuss

1. When is a relaxation’s answer useful, even though it can’t be used as it is?
2. Why is INFEASIBLE good news for a modeller?
3. What does a responsible modeller owe the people affected by a plan?

Check yourself

- Name one problem from your own life, the tool you'd use, and the one number you'd report besides the answer.